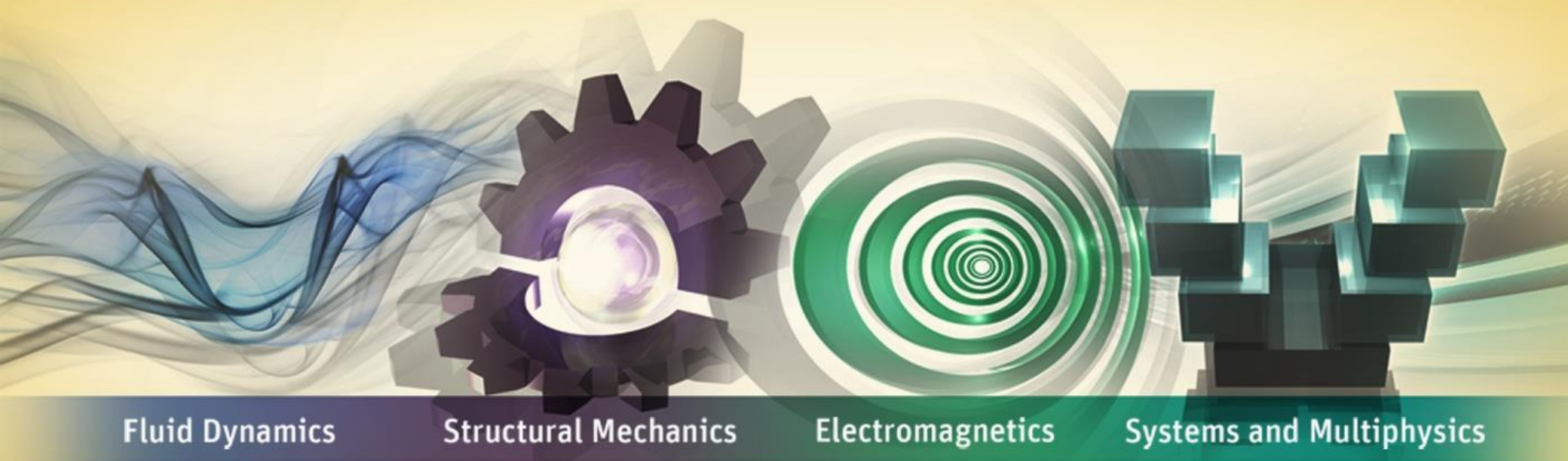


Workshop 7A: Hyperelastic Curve Fitting



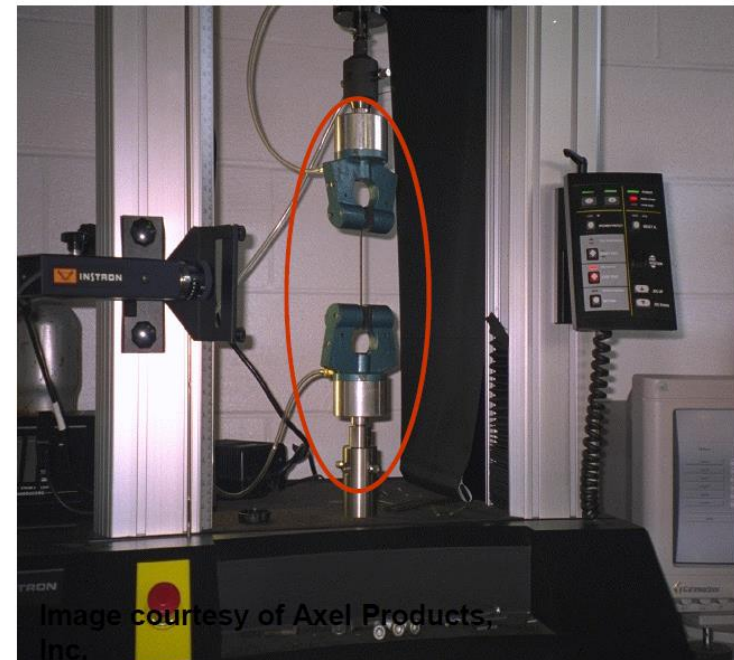
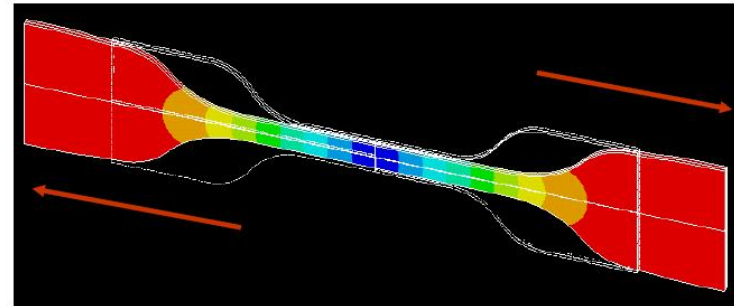
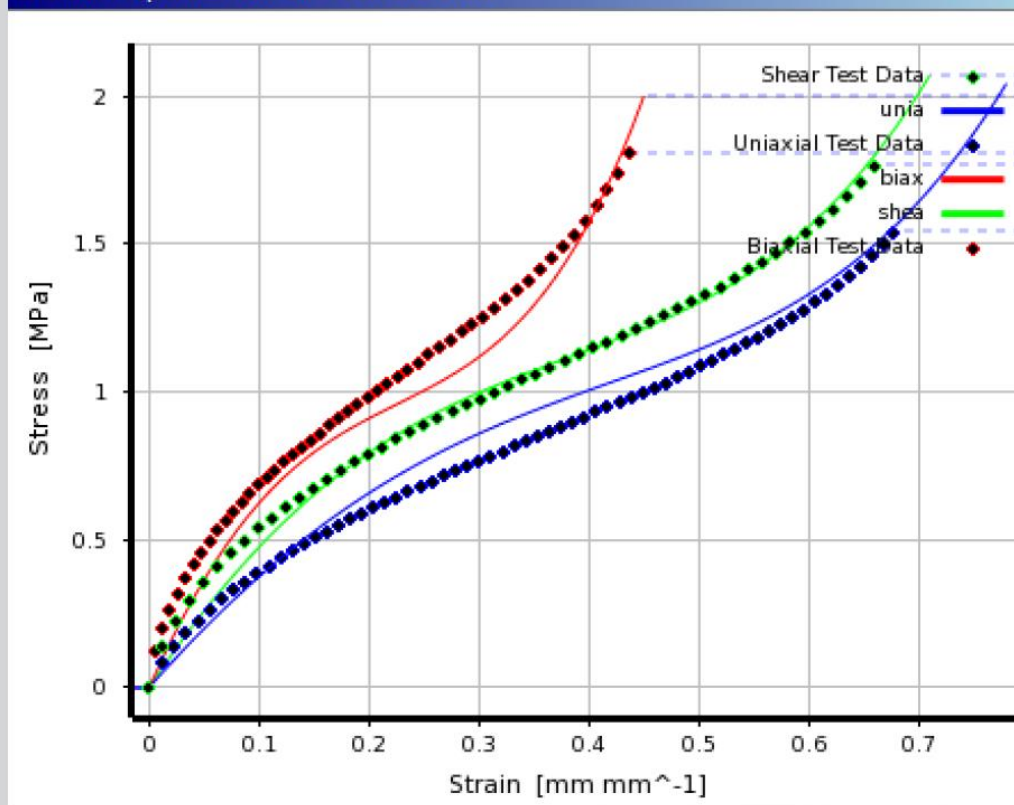
ANSYS Mechanical Introduction to Structural Nonlinearities

Workshop 7A - Hyperelastic Curve Fitting

Goal

- Use curve-fitting tool to create a hyperelastic material model from test data.
- Analyze 3d tensile rubber test specimen
- Plot and graph results

Chart of Properties Row 30: Yeoh 3rd Order

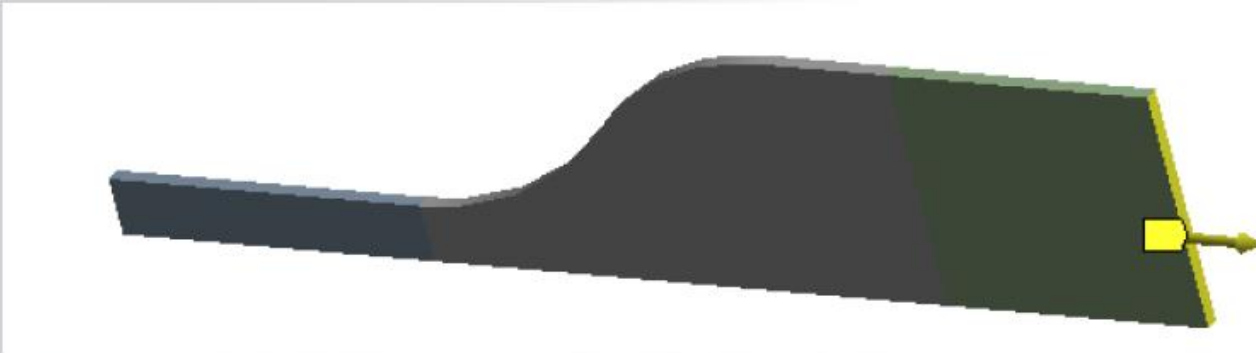


Model Description

- 3D large deflection with nonlinear material (hyperelastic)
- Three planes of symmetry ($1/8^{\text{th}}$ of the actual model)

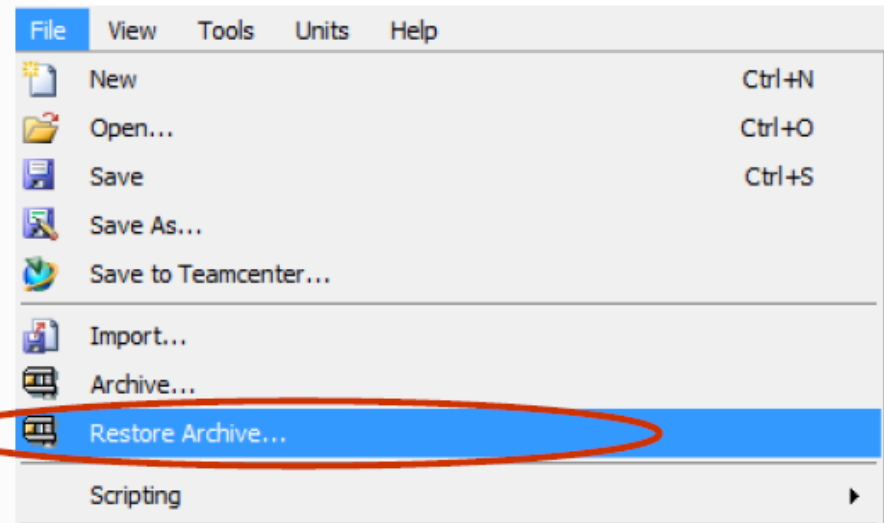
Loads and Boundary Conditions:

- Frictionless supports on each plane of symmetry
- Displacement load to elongate one end.



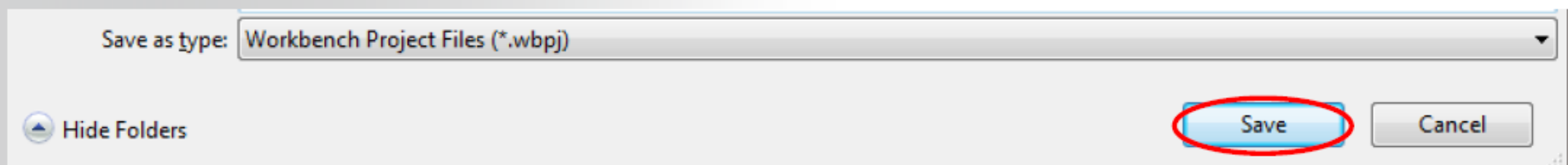
Steps to Follow:

Restore Archive... browse for file “WS4A-hyper.wbpz”



Save as

- File name: “WS4A-hyper”
- Save as type: Workbench Project Files (*.wbpj)



The project Schematic should look like the picture to the right.

- From this Schematic, you can see that Engineering (material) Data and Geometry have already been defined (green check marks).
- It remains to set up and run the FE model in Mechanical
- Highlight the Engineering Data Cell and open by clicking on the Right Mouse Button (RMB)=>Edit to verify the default linear material properties.
- Verify that the units are in Metric (Tonne,mm,...) system. If not, change this by clicking on...
 - Utility Menu=>Units=>Metric(Tonne, mm,...)

Project Schematic

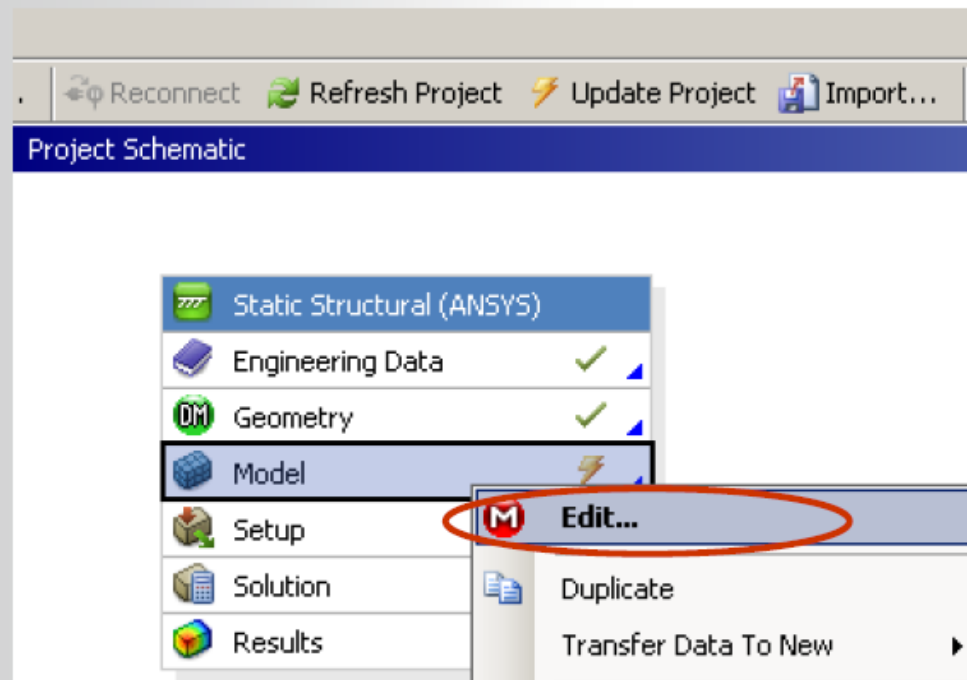
A		
1	Static Structural (ANSYS)	
2	Engineering Data	✓
3	Geometry	✓
4	Model	⚡
5	Setup	⚡
6	Solution	⚡
7	Results	⚡

Static Structural (ANSYS)

Properties of Outline Row 3: Structural Steel			
	A	B	C
1	Property	Value	Unit
2	Density	7.85E-09	tonne mm ⁻³
3	Coefficient of Thermal Expansion		
4	Coefficient of Thermal Expansion	1.2E-05	C ⁻¹
5	Reference Temperature	22	C
6	Isotropic Elasticity		
7	Young's Modulus	2E+05	MPa
8	Poisson's Ratio	0.3	
9	Alternating Stress Mean Stress	Tabular	
10	Scale	1	

Click on 'Project Tab'

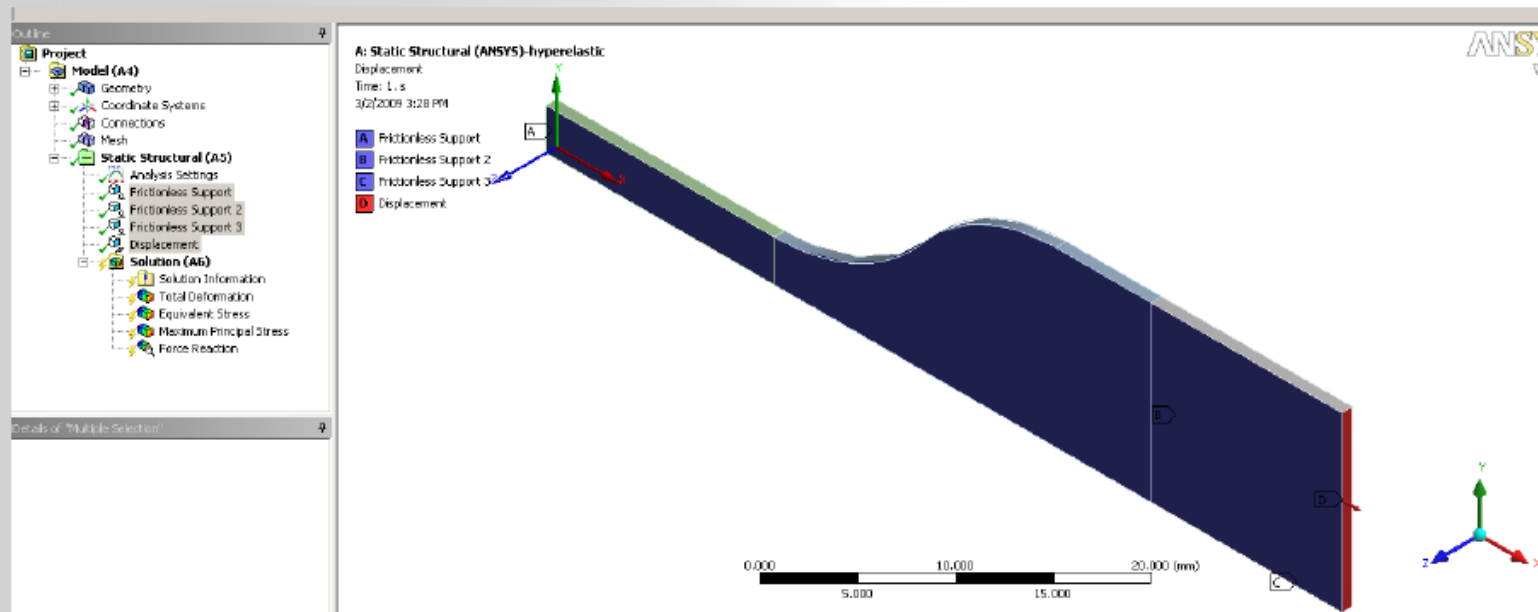
Double click on the Model Cell to open the FE Model (Mechanical Session) (or
RMB=>Edit...)



... Workshop 7A - Hyperelastic Curve Fitting

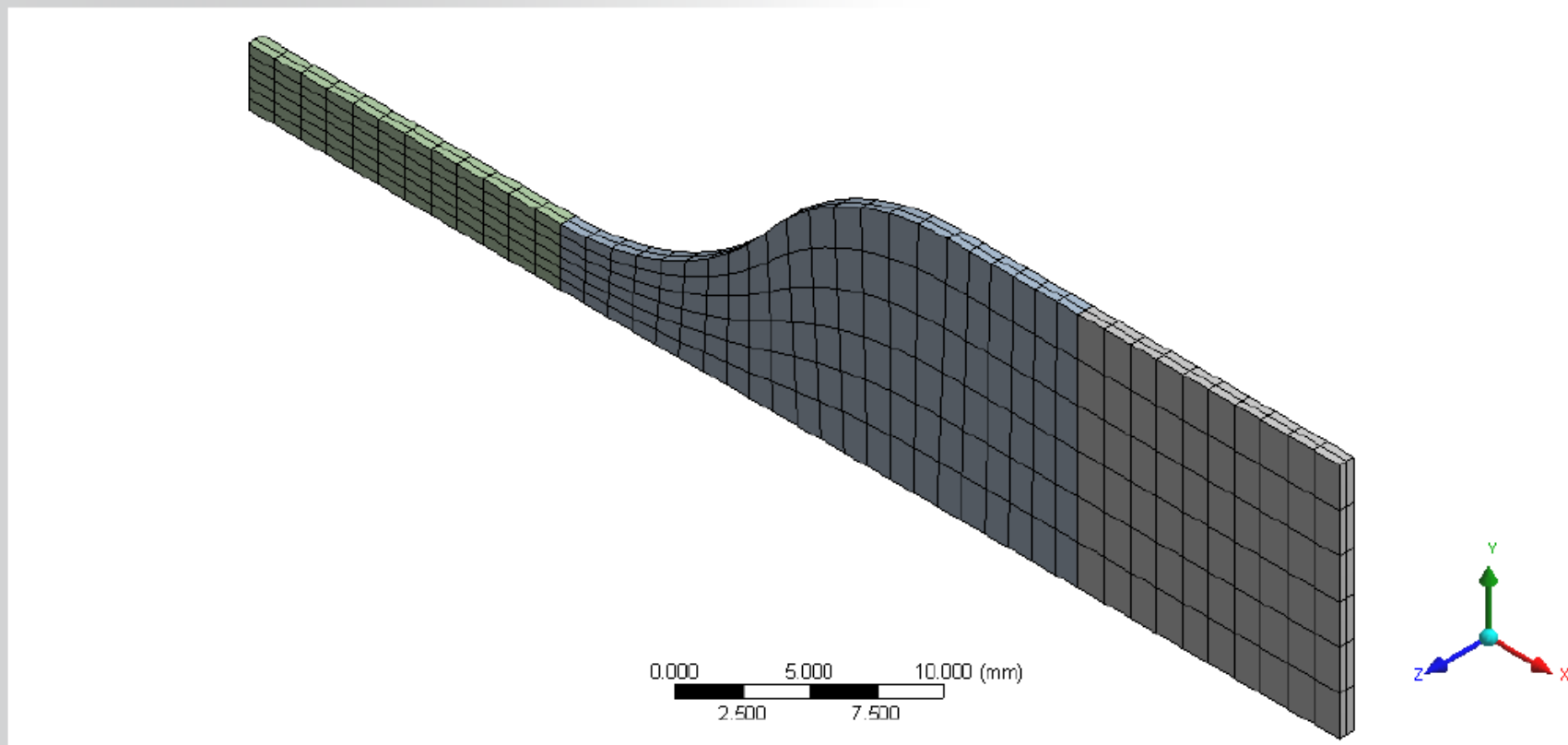
Open the folders beneath the model branch to become familiar with the model set-up.

- Confirm the three frictionless boundary conditions.
- Confirm the displacement load of 19mm on the far end.



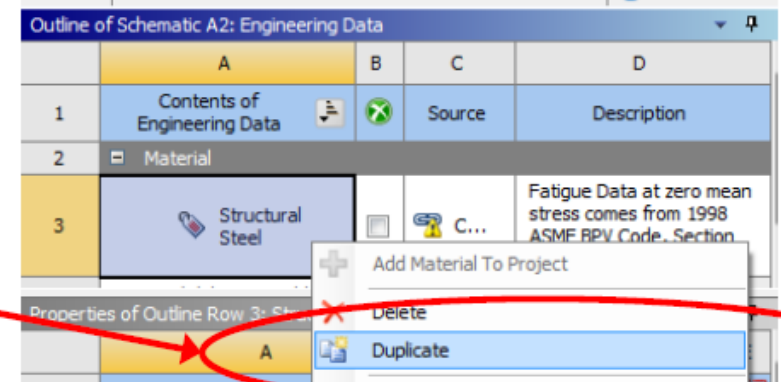
Preview the mesh:

The default mesh should be swept with hexahedral elements



Introduce hyperelastic material test data

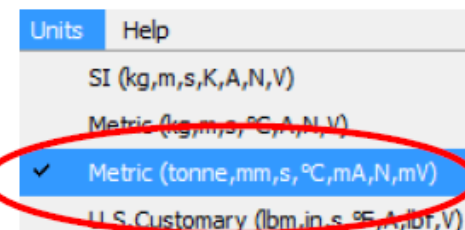
- Return to the Project Schematic
- Open the Engineering Data Cell
- Highlight the “Structural Steel” material and RMB duplicate.



- Change the name of the duplicated material to “Elastomer” and the description to “test specimen”

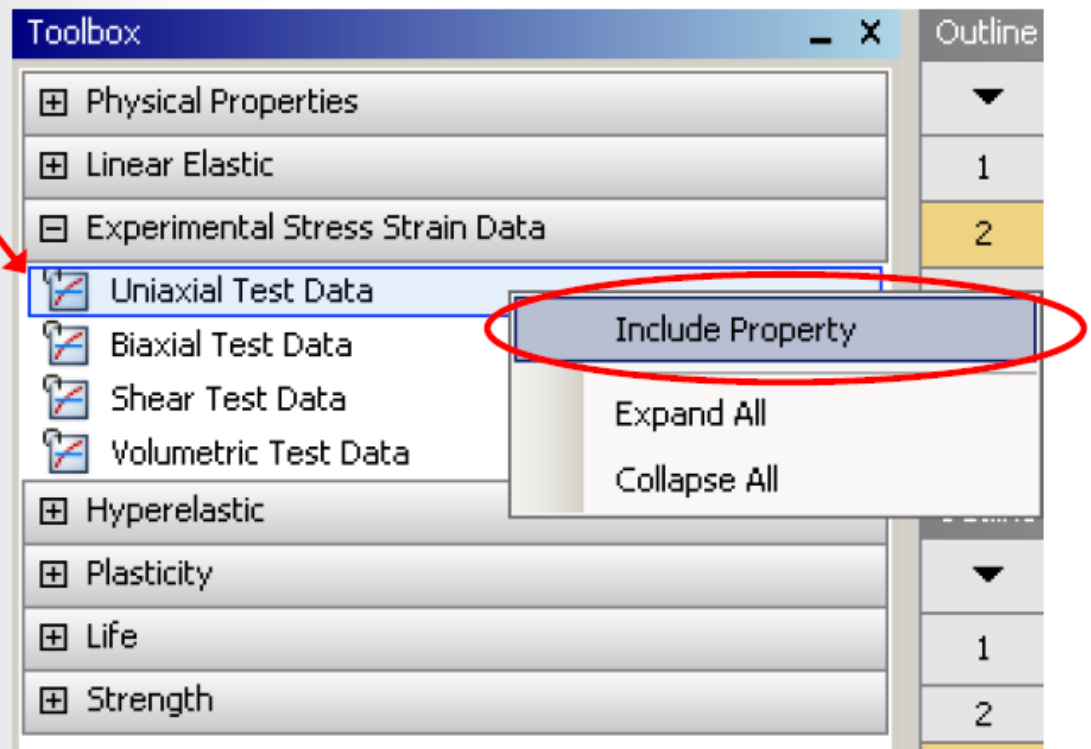


- Verify metric units are active



With the Elastomer material highlighted, insert hyperelastic material test data

- From the Toolbox, expand the “Experimental Stress Strain Data” Folder
- Highlight “Uniaxial Test Data”, RMB>Include Property
- Repeat for:
 - Biaxial Test Data
 - Shear Test Data

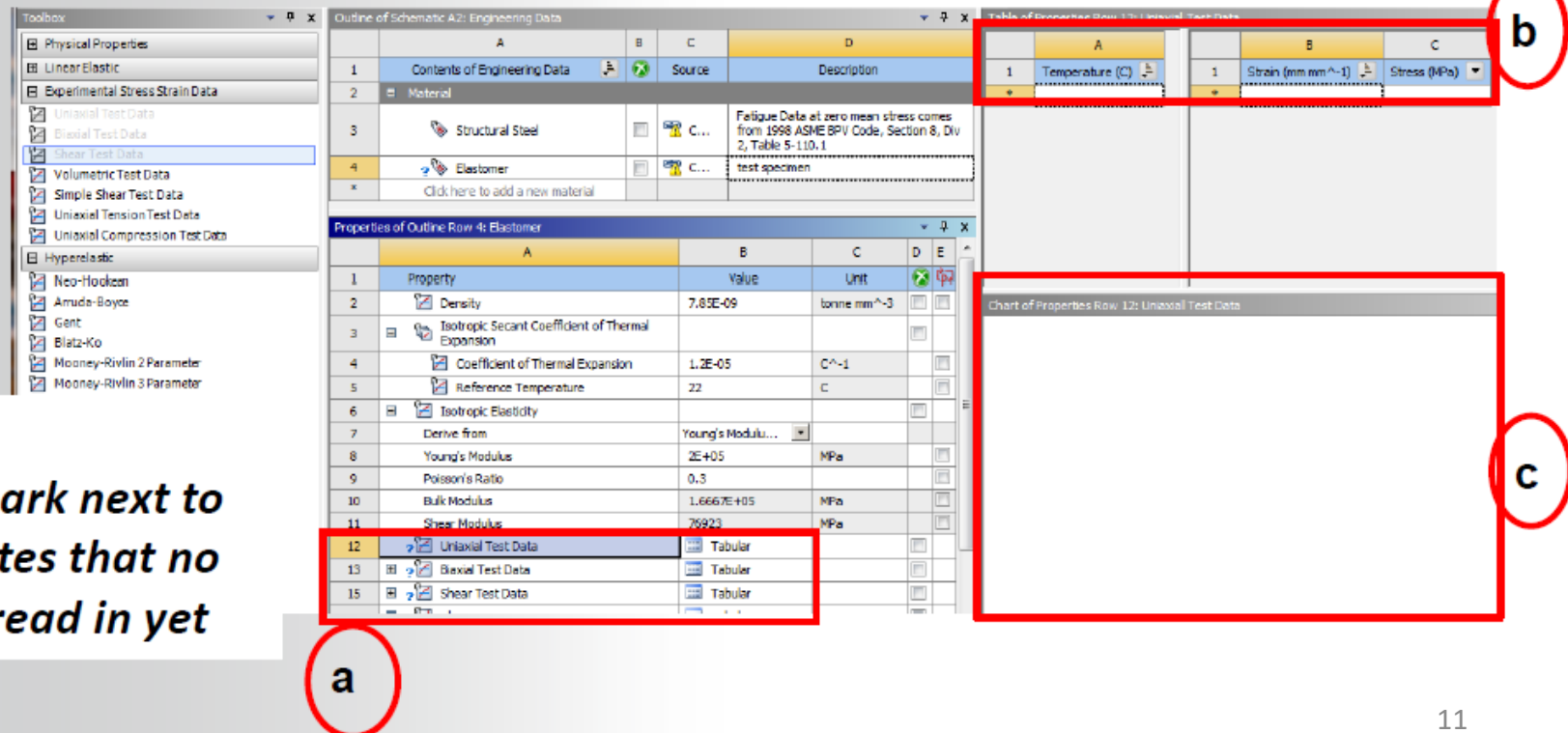


... Workshop 7A - Hyperelastic Curve Fitting

Become familiar with the property tables and chart

- Properties Outline Dialogue box should now include three additional cells for the hyperelastic test data
- The Table of Properties represents the data for the cell that is highlighted in "a".
- The Engineering Data Chart is a graphical display of the tabulated data in "b".

Note:  The question mark next to each cell indicates that no data has been read in yet

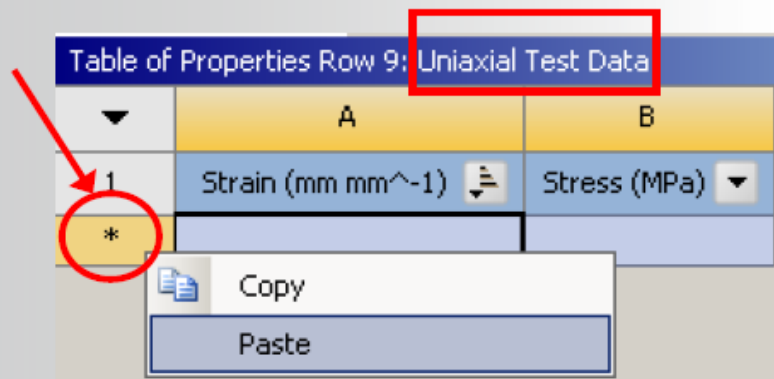


The screenshot displays the ANSYS Engineering Data interface. On the left is the 'Toolbox' with categories like Physical Properties, Experimental Stress Strain Data, and Hyperelastic. The 'Outline of Schematic A2: Engineering Data' window shows a table of properties for 'Elastomer'. The 'Table of Properties Row 12: Uniaxial Test Data' window is highlighted with a red box and labeled 'b'. The 'Chart of Properties Row 12: Uniaxial Test Data' window is highlighted with a red box and labeled 'c'. The 'Properties of Outline Row 4: Elastomer' window shows a table of properties for 'Elastomer' with columns A, B, C, D, and E. The table includes properties like Density, Isotropic Secant Coefficient of Thermal Expansion, Coefficient of Thermal Expansion, Reference Temperature, Isotropic Elasticity, and Young's Modulus. The 'Uniaxial Test Data' row is highlighted with a red box and labeled 'a'.

Property	Value	Unit
Density	7.85E-09	tonne mm ⁻³
Isotropic Secant Coefficient of Thermal Expansion		
Coefficient of Thermal Expansion	1.2E-05	C ⁻¹
Reference Temperature	22	C
Isotropic Elasticity		
Derive from	Young's Modulus	
Young's Modulus	2E+05	MPa
Poisson's Ratio	0.3	
Bulk Modulus	1.6667E+05	MPa
Shear Modulus	76923	MPa
Uniaxial Test Data	Tabular	
Bi-axial Test Data	Tabular	
Shear Test Data	Tabular	

Read in material test data

- Browse for and open “tension-uni.xls”
 - This spread sheet contains two columns of data that represent uniaxial strain-stress data
- Highlight the two columns and copy them with ‘Ctrl-C’
- Return to the “Table of Properties Row 9: Uniaxial Test Data” in the Engineering Data Cell
 - Highlight the asterisk * at the bottom of the Uniaxial Test Data Table and RMB>Paste to the insert data



Microsoft Excel - tension-uni.xls

	A	B	C
1	0	0	
2	0.011665	0.085112	
3	0.022715	0.140359	
4	0.033918	0.188141	
5	0.045122	0.22796	
6	0.055711	0.264045	
7	0.066454	0.299633	
8	0.077351	0.329993	
9	0.088094	0.355	
...	...	...	...
57	0.587036	1.255269	
58	0.597319	1.280156	
59	0.607448	1.305541	
60	0.617577	1.332915	
61	0.628013	1.361784	
62	0.638143	1.392891	
63	0.648272	1.424994	
64	0.658401	1.460831	
65	0.66853	1.499653	
66	0.67866	1.54196	

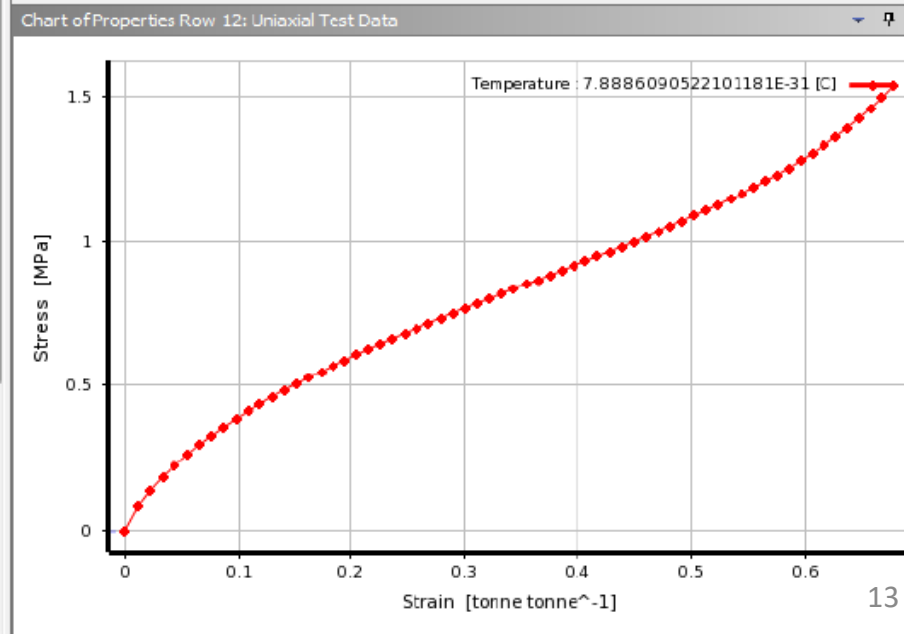
... Workshop 7A - Hyperelastic Curve Fitting

The Table of Properties should now display the Uniaxial Test Data in tabular form.
The Chart should display of the data in graphical form.

Outline of Schematic A2: Engineering Data			
	A	B	C
1	Contents of Engineering Data	Source	Description
2	Material		
3	Structural Steel	C...	Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table 5-110.1
4	Elastomer	C...	test specimen
*	Click here to add a new material		

Properties of Chart : Uniaxial Test Data Elastomer				
	A	B	C	D
1	Property	Value	Unit	
2	Density	7.85E-09	tonne mm ⁻³	
3	Isotropic Secant Coefficient of Thermal Expansion			
4	Coefficient of Thermal Expansion	1.2E-05	C ⁻¹	
5	Reference Temperature	22	C	
6	Isotropic Elasticity			
7	Derive from	Young's Modulu...		
8	Young's Modulus	2E+05	MPa	
9	Poisson's Ratio	0.3		
10	Bulk Modulus	1.6667E+05	MPa	
11	Shear Modulus	70925	MPa	
12	Uniaxial Test Data	Tabular		
15	Biaxial Test Data	Tabular		
17	Shear Test Data	Tabular		
19	Alternating Stress Mean Stress	Tabular		
20	Interpolation	Log-Log		
21	Scale	1		
22	Offset	0	MPa	
23	Strain-Life Parameters			

Table of Properties Row 12: Uniaxial Test Data		
	A	B
1	Temperature (C)	Strain (mm mm ⁻¹)
2		0
*		
3	0.011665	0.085112
4	0.022715	0.14036
5	0.033918	0.18814
6	0.045122	0.22796
7	0.055711	0.26405
8	0.066454	0.29963
9	0.077351	0.32999
10	0.088094	0.35936



... Workshop 7A - Hyperelastic Curve Fitting

Repeat procedure for biaxial test data and shear test data using excel files 'tension-eb.xls' and 'tension-pt.xls' respectively.

Table of Properties Row 12: Biaxial Test Data		
	A	B
1	Strain (mm mm ⁻¹)	Stress (MPa)
2	0	0
3	0.005654	0.12375
4	0.012419	0.20081
5	0.019286	0.26354
6	0.026454	0.32032
7	0.033725	0.37106
8	0.041096	0.41624
9	0.048164	0.45633
10	0.055535	0.49481
11	0.062805	0.53131
12	0.070277	0.56518

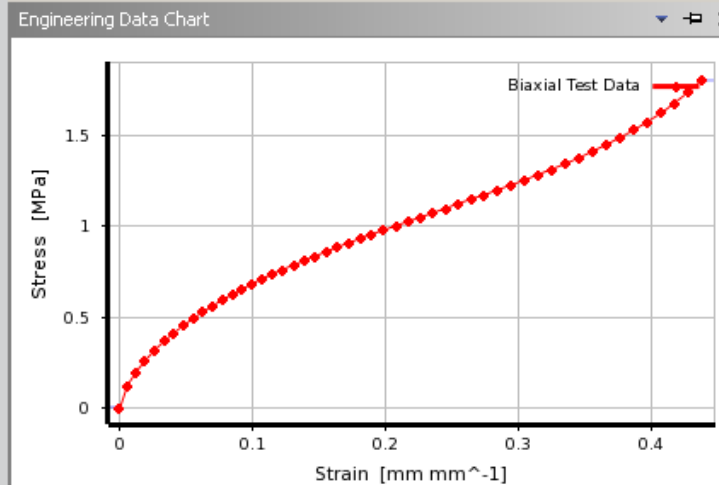
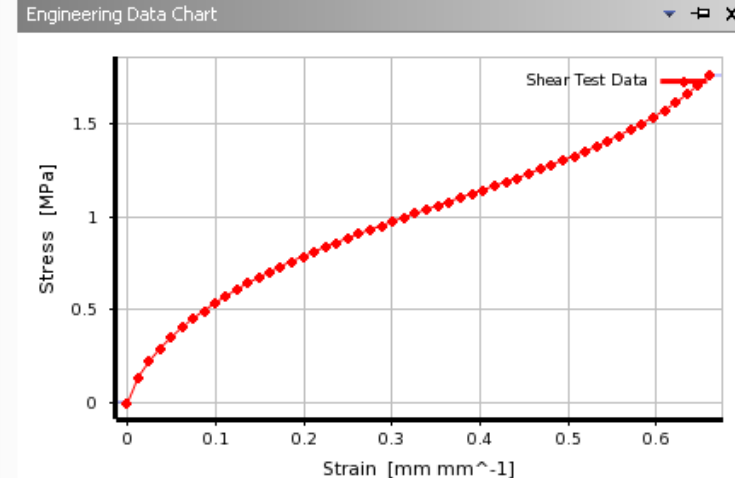


Table of Properties Row 15: Shear Test Data		
	A	B
1	Strain (mm mm ⁻¹)	Stress (MPa)
2	0	0
3	0.012065	0.13943
4	0.025226	0.22798
5	0.037657	0.297
6	0.049905	0.35608
7	0.062427	0.40869
8	0.074493	0.45613
9	0.087472	0.49853
10	0.099993	0.54031
11	0.11242	0.57764
12	0.1244	0.61315



From the Toolbox, open the Hyperelasticity Folder and highlight “Yeoh 3rd Order with RMB> Include Property.

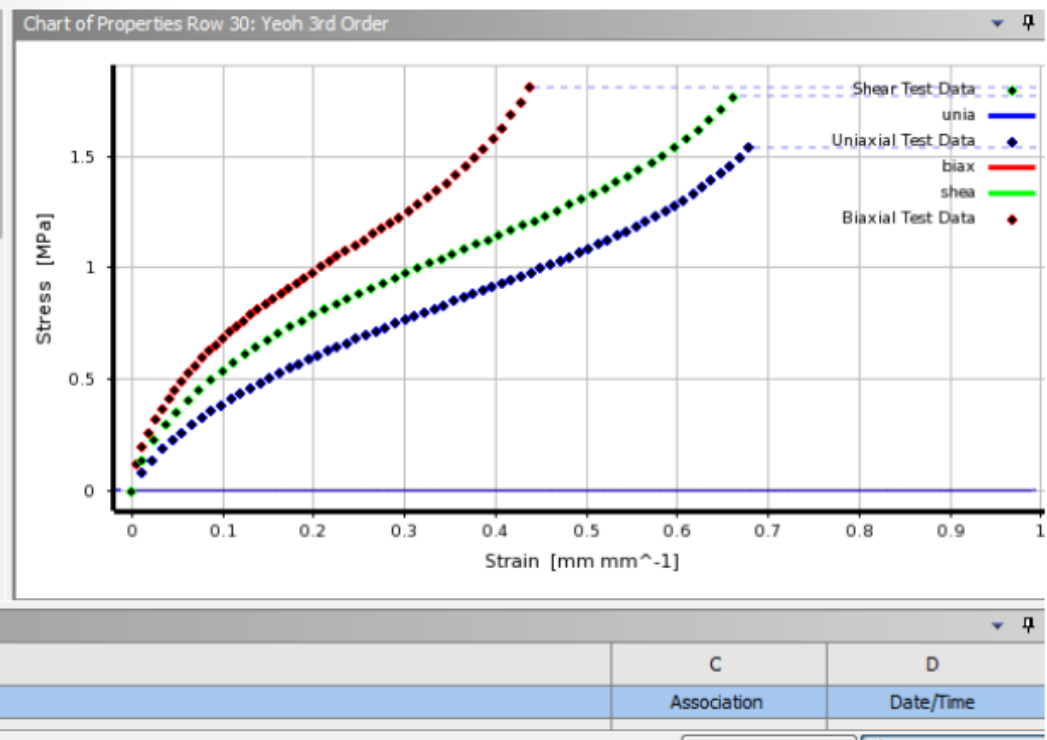
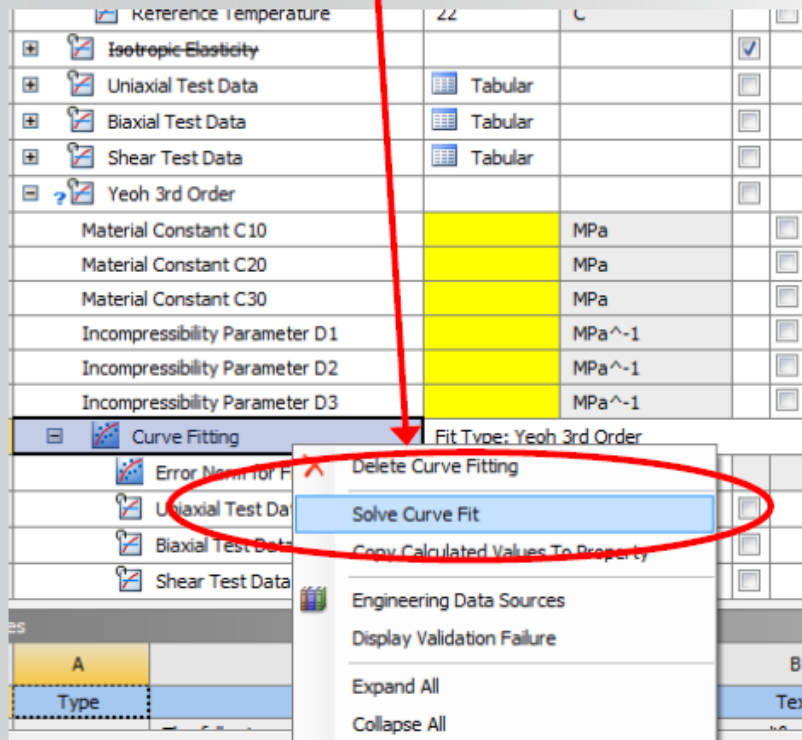
- You should now see a Yeoh 3rd order cell in the Properties Dialogue box with undefined constants; hence, the question mark.

The screenshot displays the ANSYS Hyperelasticity Properties Dialogue box. On the left, the 'Hyperelastic' folder is expanded, and the 'Yeoh 3rd Order' option is highlighted. A red circle and an arrow point to this option, and another red circle highlights the 'Include Property' button. The main table on the right, titled 'Properties of Chart : Uniaxial Test Data Elastomer', lists various material properties. The 'Yeoh 3rd Order' row (row 23) is highlighted with a red circle and contains a question mark icon, indicating that its material constants are undefined. The table also shows other properties like Bulk Modulus, Shear Modulus, and various material constants (C10, C20, C30, D1, D2, D3) with their respective units (MPa, MPa^-1).

	A	B	C	D	E
1	Property	Value	Unit		
10	Bulk Modulus	1.0007E+05	MPa		
11	Shear Modulus	76923	MPa		
12	Uniaxial Test Data	Tabular			
15	Biaxial Test Data	Tabular			
16	Has Lateral Strain	No			
17	Scale	1			
18	Offset	0	MPa		
19	Shear Test Data	Tabular			
23	Yeoh 3rd Order				
24	Material Constant C10		MPa		
25	Material Constant C20		MPa		
26	Material Constant C30		MPa		
27	Incompressibility Parameter D1		MPa^-1		
28	Incompressibility Parameter D2		MPa^-1		
29	Incompressibility Parameter D3		MPa^-1		
30	Curve Fitting	Fit Type: Yeoh 3rd Order			
31	Error Norm for Fit	Normalized Error			15
32	Uniaxial Test Data	Tabular			

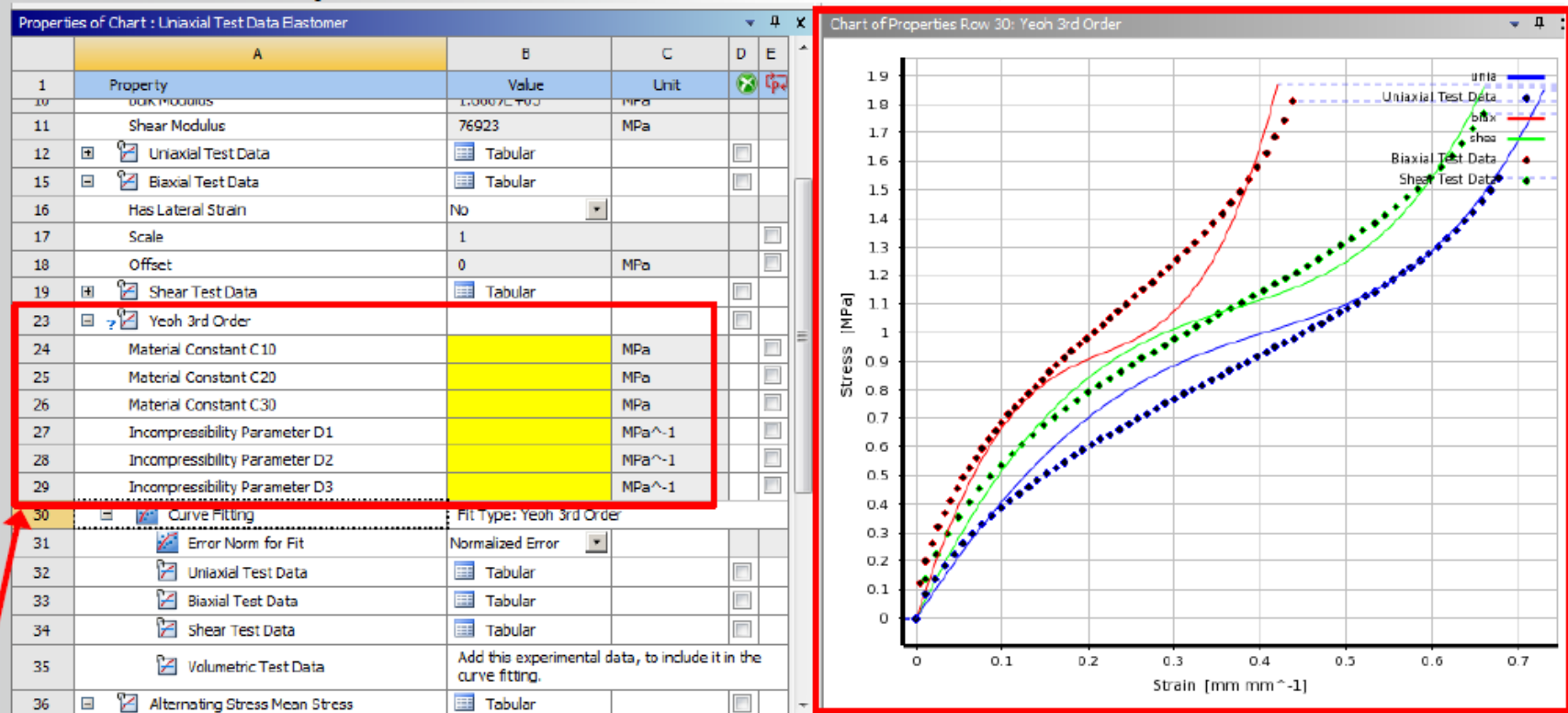
Execute "Solve Curve Fit".

- Expand the Yeoh 3rd Order Cell
- Highlight the Curve Fitting Cell
 - RMB>Solve Curve Fit



... Workshop 7A - Hyperelastic Curve Fitting

When complete, the Engineering Data Chart will display all three modes of test data in dot-dash lines. Superimposed on this is the predicted Yeoh behavior in solid lines for comparison.



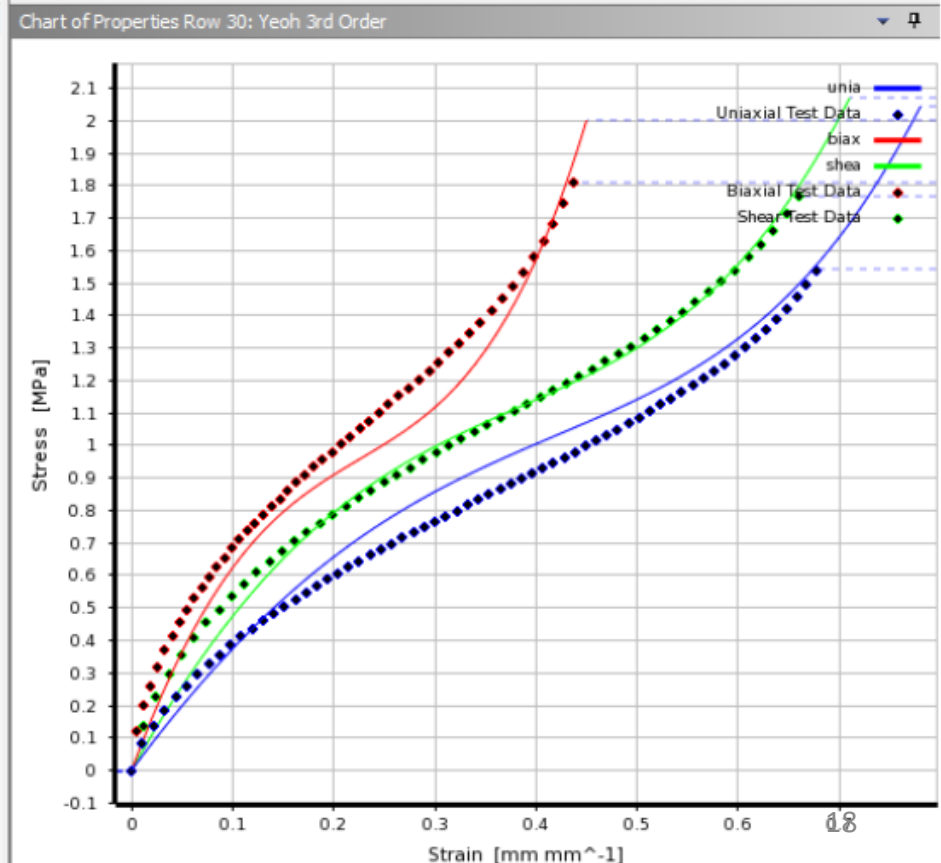
Note: The Coefficients have been calculated, but no material model has been saved to the Engineering Data yet. This is by design. If you are not satisfied with the resulting curve fit, you can try other functions, or perhaps suppress a less dominate mode of loading and rerun the curve fitting routine for a better result.

... Workshop 7A - Hyperelastic Curve Fitting

Change the Error Norm from “Normalized Error” to “Absolute Error” and run the curve fitting routine again.

- Recall the normalized error approach gives equal weight to all data points while the absolute error approach gives more weight to larger strain values.
 - This results in a slightly improved curve fit at higher strain values.

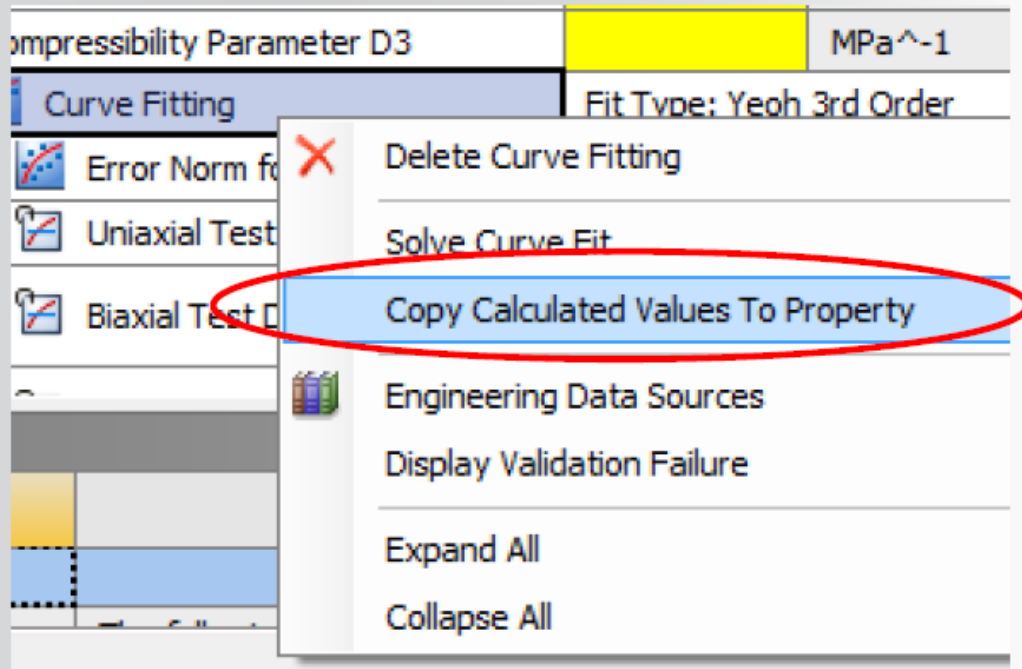
Properties of Chart : Uniaxial Test Data Elastomer				
	A	B	C	D E
1	Property	Value	Unit	
10	Bulk Modulus	1.0007E+03	MPa	
11	Shear Modulus	76923	MPa	
12	Uniaxial Test Data	Tabular		
15	Biaxial Test Data	Tabular		
16	Has Lateral Strain	No		
17	Scale	1		
18	Offset	0	MPa	
19	Shear Test Data	Tabular		
23	Yeoh 3rd Order			
24	Material Constant C10		MPa	
25	Material Constant C20	Normalized Error	MPa	
26	Material Constant C30	Absolute Error	MPa	
27	Incompressibility Parameter D1		MPa ⁻¹	
28	Incompressibility Parameter D2		MPa ⁻¹	
29	Incompressibility Parameter D3		MPa ⁻¹	
30	Curve Fitting	Fit Type: Yeoh 3rd Order		
31	Error Norm for Fit	Absolute Error		
32	Uniaxial Test Data	Tabular		
33	Biaxial Test Data	Tabular		
34	Shear Test Data	Tabular		
35	Volumetric Test Data	Add this experimental data, to include it in the curve fitting.		



... Workshop 7A - Hyperelastic Curve Fitting

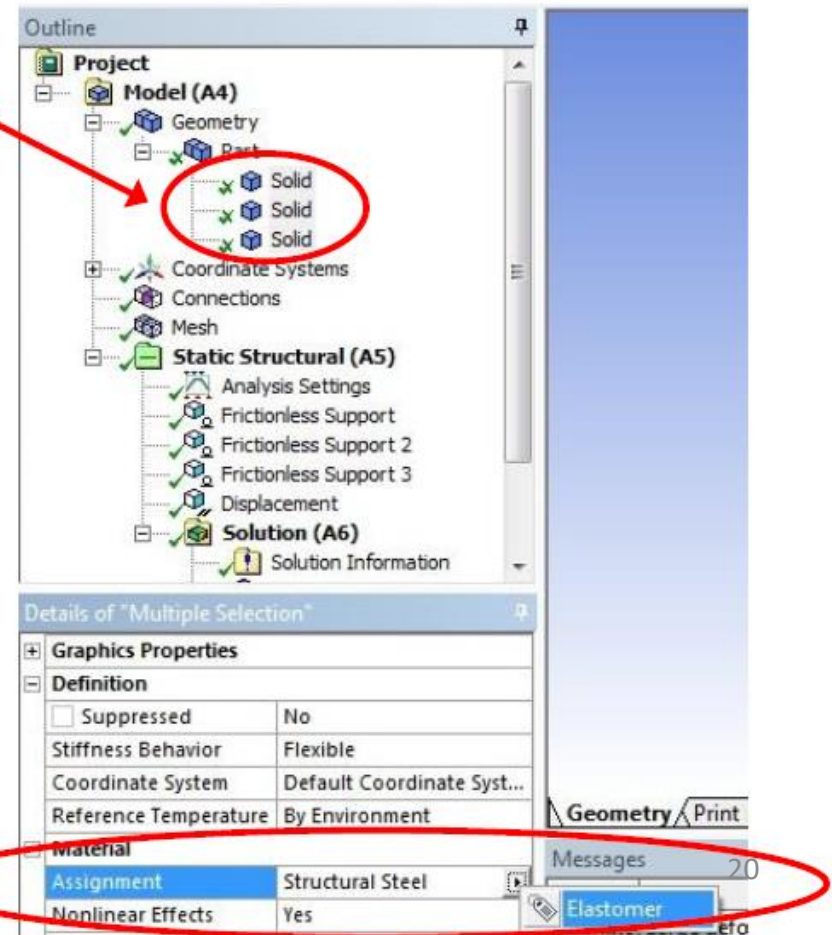
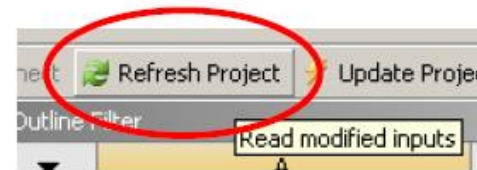
Once a successful curve fit is achieved, copy the data to the Engineering properties

- Highlight “Curve Fitting” RMB> “Copy Calculated Values to Property”
- The Yeoh coefficients are now copied to the Properties Table. Notice the “D” parameters are all zero. This is in keeping with the assumption that the material will be fully incompressible if no volumetric strain data is read into the curve fitter.



Properties of Chart : Uniaxial Test Data Elastomer			
	A	B	C
1	Property	Value	Unit
23	Yeoh 3rd Order		
24	Material Constant C10	0.6976	MPa
25	Material Constant C20	-0.24484	MPa
26	Material Constant C30	0.12629	MPa
27	Incompressibility Parameter D1	0	MPa ⁻¹
28	Incompressibility Parameter D2	0	MPa ⁻¹
29	Incompressibility Parameter D3	0	MPa ⁻¹

- Click on 'Project Tab'
- From the Utility Menu:
 - Click on "Refresh Project"
- Return to the Mechanical Session and highlight the 3 bodies that make up this part and change the default material assignment from "Structural Steel" to "Elastomer"



... Workshop 7A - Hyperelastic Curve Fitting

Return to the Mechanical Session and specify Analysis Settings as shown below:

Turn on Auto Time Stepping →

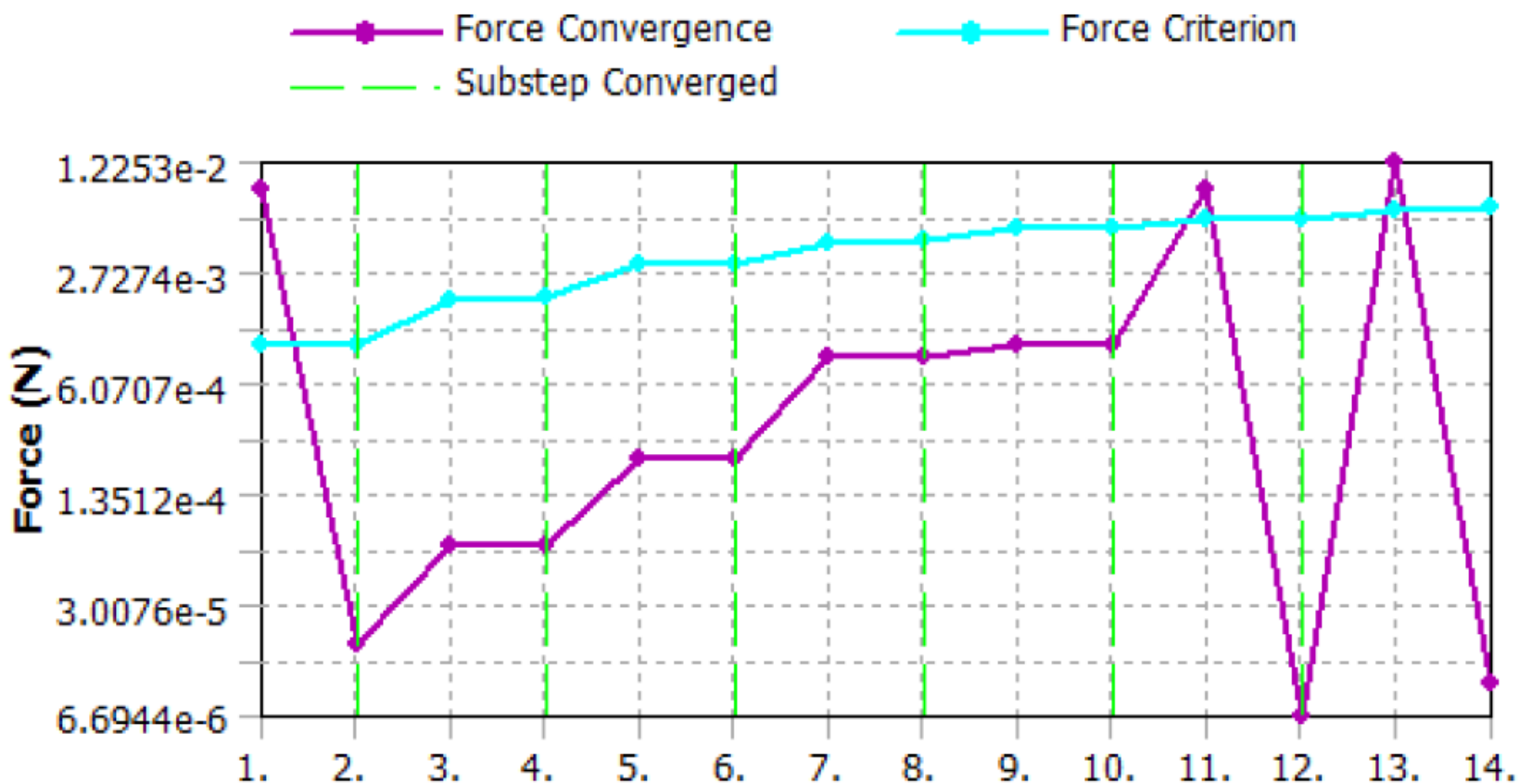
Specify enough substeps
to enhance convergence →

Turn on Large Deflection →

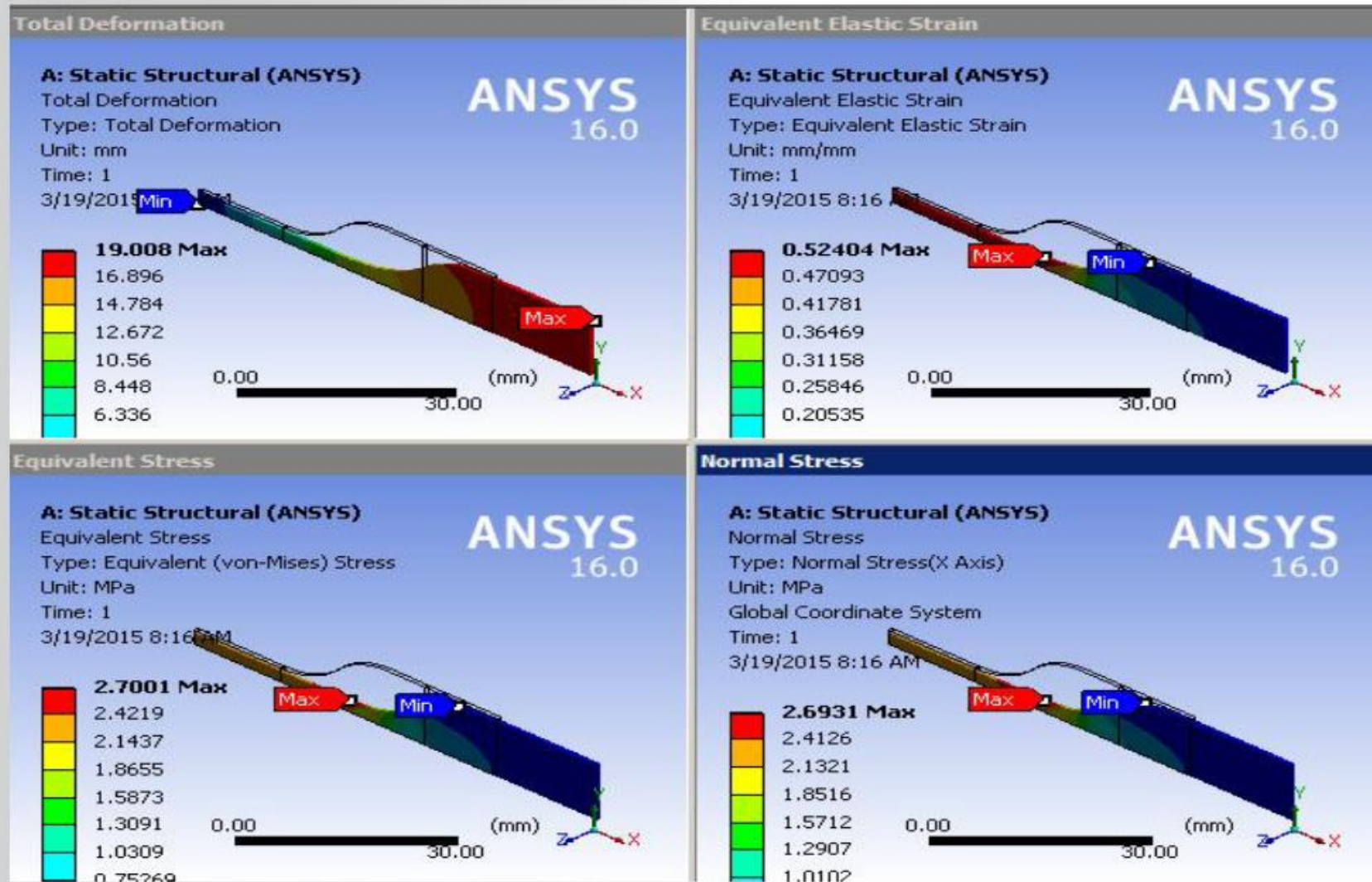
Details of "Analysis Settings"	
[-] Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	10.
Minimum Substeps	5.
Maximum Substeps	50.
[-] Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Large Deflection	On
Inertia Relief	Off
[+] Restart Controls	
[+] Nonlinear Controls	
[+] Output Controls	

- Execute the Solve

Model converges very nicely in less than 20 iterations

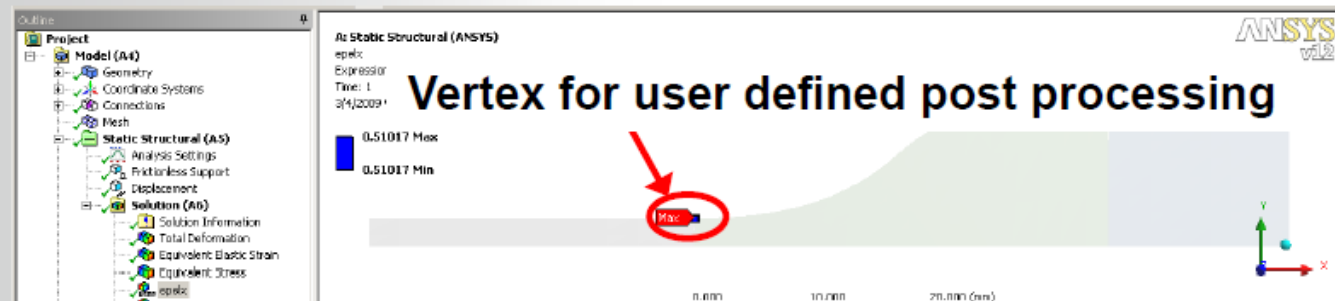


Post process results (i.e. deformation, stress, strain,...etc).



... Workshop 7A - Hyperelastic Curve Fitting

Select Vertex as shown and insert User defined results to post process stress and strain along x direction

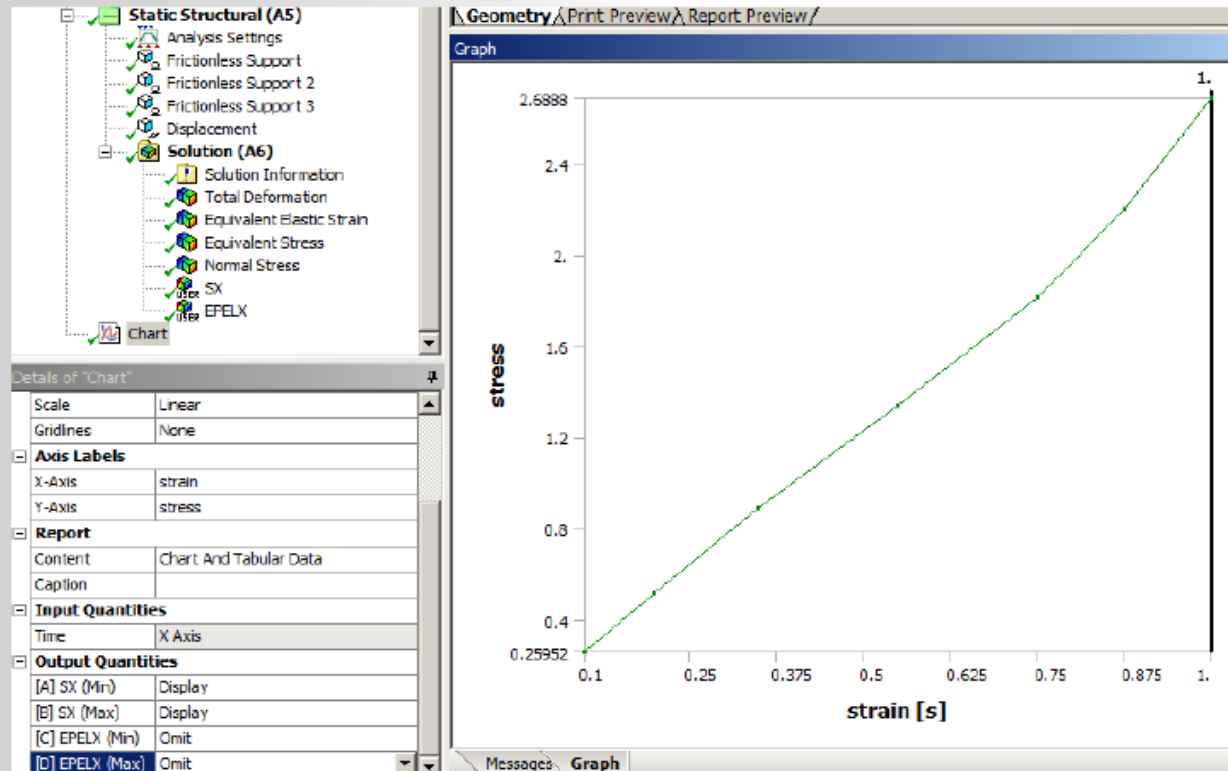


- Expression for x direction of stress is “SX”
- Expression for x direction of strain is “EPELX”
 - Refer to documentation on PRNSOL command for syntax of user defined results

Valid item and component labels for element results are:

S	X, Y, Z, XY, YZ, XZ	Component stress
"	1, 2, 3	Principal stress
"	INT	Stress intensity
"	EQV	Equivalent stress
"	MAXF	Maximum Stress Failure Criteria [2][7]
"	TWSI	Tsai-Wu Strength Index Failure Criterion [2][7]
"	TWSR	Inverse of Tsai-Wu Strength Ratio Index Failure Criterion [2][7]
EPEL	X, Y, Z, XY, YZ, XZ	Component elastic strain
"	1, 2, 3	Principal elastic strain

Add chart tool and insert EPELX and SX user defined results as objects for plotting stress vs strain as shown below.

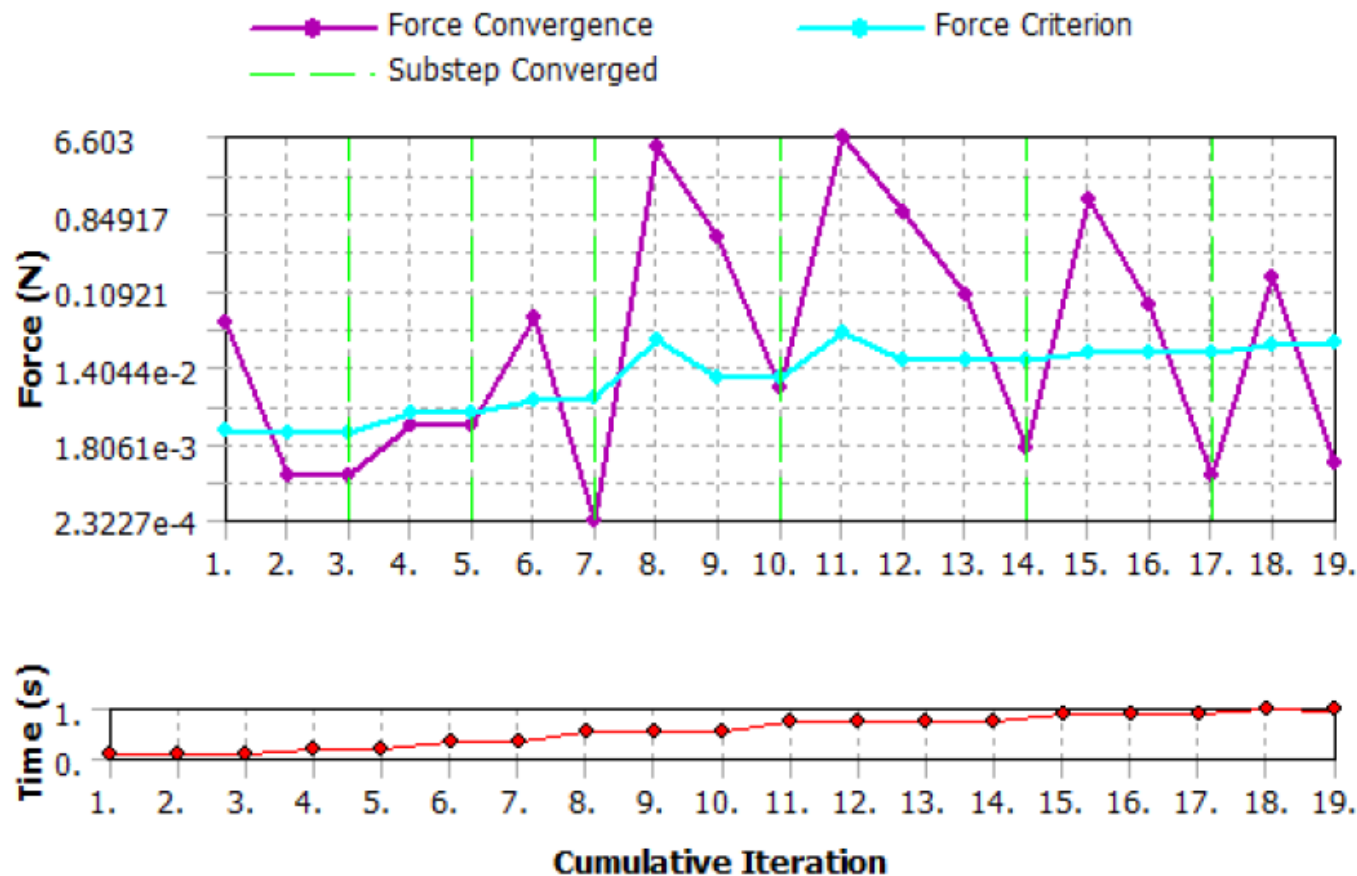


Q. Why doesn't this curve match up exactly with the tensile test data curve?

A. The test data is engineering stress-strain, the results are post processed as true stress-strain.

Change the displacement load to 50mm and re-run the solution.

- Model still converges without trouble in less than 20 iterations, no bisections, even though the loading is outside of the bounds of the original test data.



Notice the dramatic stress stiffening effects at higher strain values, a common characteristic of most elastomers.

